

A comparative study of Different Denoising Techniques in Digital Image Processing

Kamran Ahmad
School of Electronics & Information
Engineering,
Tianjin Polytechnic University
Tianjin, 300387, China
kamran_mani25@yahoo.com

Jawad Khan
School of NARI Electric & Automation
Nanjing Normal University, Nanjing
China 210042
Jawadkhans@outlook.com

Muhammad Salah Ud Din Iqbal
School of computer science
Bahaudin Zakariya University,
Multan, Sub-campus Sahiwal, Pakistan
salahudiniqbal@gmail.com

Abstract— The quality of image usually depends on various factors including noise, light, and temperature. Noise has a decisive role in the image as it is constantly present in different digital images. Noise implication may present during the image coding, transmission and development process. Noise removal has become an eye-catching and dynamic field in the Image processing domain. There is a significant recent advance in filtering of different kinds of noises for digital image processing. In this paper the problem of de-blurring in the presence of various noises is studied. A comparison among various filtering techniques is made in the presence of different kinds of noises. The experiments show that the Adaptive filter method performance is appealing both visually and in terms of objective quality measures that as the color peak-signal-to-noise ratio (CPSNR) and the mean square error (MSE) for Gaussian noise, Speckle noise, Poisson noise, while median filter demonstrates the visual objectives for salt and pepper noise.

Keywords- Image Processing, Noise Types, Noise Removal Filter, Gaussian Noise, Salt and Pepper Noise.

I. INTRODUCTION

The quality of the digital images depends on various factors. The image tends to have a good quality, when it has minimum disturbing factors such as salt and pepper noise, Gaussian noise and mixed noise etc. Noise can be considered as any deprivation in the image signal, instigated by exterior disturbance [1]. Noise is one the most crucial factors for the degradation of the quality of the image. The attainment procedure for digital images changes optical signals into electrical signals. Furthermore, they are also converted into digital signals and later process according to the noise is presented in digital images.

When an image is transformed from one form to another that is the scanning, transmitting, digitizing storing etc. There for the Image can be disturbed by numerous ways of transporting image from one place to another. This will include electronically transmission of an image through wireless network or satellite. One might anticipate errors while receiving these images. These errors are incurred on the result of the output image [2] [3]. The errors on the images depend on the type of the disturbance in the images. Therefore, it is becoming very critical to have noise removal techniques.

Noise can be found in numerous forms in the image. The most popular noise or in other words the disturbing factors, such as salt and pepper noise (Impulse noise) has the black and white pixels on the images. A compelling noise removing

strategy for this sort of noise includes the utilization of middle channel or contrast symphonious middle channel [4] [5] [6]. Another popular form of noise is Gaussian noise, possessing an approximation density function equivalent to that of the normal distribution, which is likewise recognized as the Gaussian distribution. Gaussian noise can also be explained as the values that the noise can take on are Gaussian distributed [7] [8] [9] [10]. Kaur *et al.* claim that the Gaussian noise can be eliminated by using modified adaptive bilateral filter [11].

Numerous noise removal techniques have been used by enormous researchers in the recent years. In order to completely eliminate the noise from the digital image we compared the most robust and efficient filtering techniques and the noise removing algorithms to accomplish effective results without disturbing the integrity of the image.

There are numerous researchers which have explored the field and have done experiments on different images. In 2008, Tripathi and Lipton, using Generative Adversarial Network (GAN) have proposed correction by projection and de-noising [12]. They suggested de-noising the corrupted images and achieved to de-noise the images with the help of exploration of the nearest point Generative adversarial networks. The author de-noise the corrupted image in two phases i.e. by approximately recuperate the latent vector, therefore obtaining advanced quality image and associating with BM3D. Secondly, we establish the latent vectors recuperated from noisy images exhibit a reliable bias.

In 2015, J.M *et al.* [13] proposed another noise removal method but he performed his experiments on grey scale. Noise pixel modification method was which were also based on two phases. In the first phase they used fuzzy method and mixed noise filtering step. The noises removed from the image in this research, were a combination of Gaussian noise and as well as salt and pepper noise. This method had outperformed many fuzzy logics while balancing tradeoff between noise and detail preservation. In the same year M. Afifi *et al.* [14] suggested a modified Poisson mix technique. Their experiment results were encouraging as MPB (Modified Poisson Blending) moderates the bleeding problems and produces more natural composite images as compared to other techniques. Rather than employing pixel powers, they use the angle area operated as a part of Poisson picture progress, it suffers from two key matters, and first one is color reflection and second on is reflection objects.

Patel *et al.* [15] proposed the combination grey level clustering and contrast development algorithm. Their motive was to improve compare features and defects of noise. This

practice is very supportive to visualize breast tumors in higher density that additionally benefits in recognition of breast cancer. The breast mammograms were improved without any forfeiture of image artifacts. This process is simple and computationally productive. Furthermore, the process maintained the mean brightness, despite the fact that preserving the attributes of the image and further avoids any un-wanted attributes that might occur during the conventional process [18].

An effective method to remove the Gaussian noise was proposed by Yang *et al.* [16]. In this method, they proposed an effective de-noising algorithm that is a combination of Poisson and Gaussian probability distribution, to construct a more precise and implement the contour-let transform which delivers a sparse demonstration of the directional constituents in images. They also include the hidden Markov models with a framework and noise approximation in the transform domain. The experiment result shows that the improved images after the implementation of the proposed image.

We investigate and compare the introduction and removal of noise to the digital image. We have indulged different types of noise and further studied the impact on the digital images because of these noises. We have developed an algorithm to introduce these noises to the digital images. These noises include “Salt and Pepper Noise”, “Gaussian Noise”, “Photon Noise”, and “Speckle Noise”.

We have also investigated the filters that are used to remove the noise from the images. Further discussion about filtering techniques has been done in the Section 3. We will use these filters in the later stages of our experiments to remove our applied noises. We have used four filters for our experiments i.e. “Median Filter”, “Linear Filter”, “FIR Filter”, “Adaptive filter”. Our methodology based on the insight examination of the De-noising field of the image processing. The rest of the paper is organized as follows. The second section of this paper presents noise types, the third section deals with filtering techniques, the fourth section methodology, the fifth section results and discussion, and finally conclusion and future work are discussed in the sixth section of the paper.

II. NOISES AND THEIR IMPACT ON IMAGE

Following are the types of noises that we have investigated during our research. We have incorporated these noises to HD (High Definition) quality color image and try to figure out the impact of these applied noises.

A. Salt and Pepper Noise

Unexpected or sharp disruption may cause noise, which is produced in the image signal. The scattered pixels (combination of white and black: the reason why it is called Salt & Pepper) generated, over the image. Djurović *et al.* [17] presented BM3D filter to remove salt and pepper noise. Salt and Pepper noise disturb the image in a scenario when there is a transformation of an image from one place to another. This inclined the image to form as disturbed image. Salt and pepper model have only two possible values, a and b and the probability of locating each of them is less than 0.1. For 8 bit pixel image, normally intensity value for pepper noise is close to 0 and for salt noises its 255.

$$PDF_{salt\&pepper} = \begin{cases} A & \text{for } g = a(\text{"pepper"}) \\ B & \text{for } g = b(\text{"salt"}) \end{cases}$$

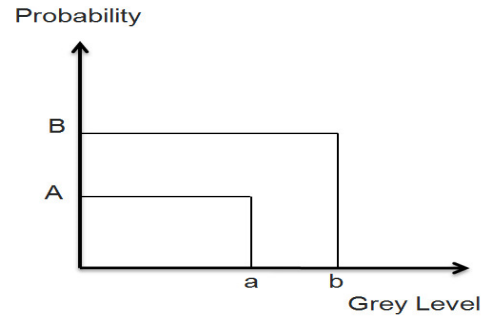


Fig. 1. Probability Density function for salt and pepper noise model

B. Gaussian Noise

Some scholars called Gaussian noise as the additive noise. Gaussian noise is produced by the haphazard oscillation in the signals. The probability density function P of Gaussian random variable (z) can be calculated as follows:

$$P_G(Z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

Where z represents the grey level, μ the mean value and σ the standard deviation.

C. Poisson Noise

This noise is generated by the unsystematic variations of photons. The researchers also called the Poisson noise as photo shot noise, due to the impact of the photons; it enters one sensor then the other.

D. Speckle Noise

Speckle noise also known as multiplicative noise as the noise can be modeled by random values multiplied by pixel values; hence it is called multiplicative noise. It causes major problem in some radar applications. Its probability function follows gamma radiation which can be calculated by following equation.

$$F(g) = \frac{g^{\alpha-1} e^{-\frac{g}{a}}}{\alpha - 1! a^\alpha}$$

III. FILTERING TECHNIQUES

A. Adaptive Filter

It is commonly used to enhance or restore data by removing noise without significantly blurring the structure in the image.

$$\hat{f}(x, y) = g(x, y) - (\sigma_\eta^2 / \sigma_i^2) [g(x, y) - m_i]$$

Where σ_i^2 – local variance of local region
 m_i – local mean

σ_η^2 – variance of overall noise

$g(x, y)$ – pixel value at position (x, y)

B. Median Filter

Median filter is nonlinear filtering technique. It is widely used to remove salt and pepper noise and preserves the edges. The median filter calculates by first sorting all the pixel values from window into numerical order, and then replacing the pixel being considered with the middle (median) pixel value.

C. FIR Filter

FIR filter has symmetrical impulse response in the spatial domain (region of support centered about the origin) which offer a significant reduction in number of multiplications necessary for filter realization. Advantages of FIR filter is it's always stable.

D. Linear Filter

Linear filter in which the value of an output pixel is linear combination of the values of the pixels in the input pixel neighborhood. Disadvantages of this filter that noise become increase in it.

IV. METHODOLOGY

Our experiment's methodology consists of three steps, i.e. first step; consists of the introduction of the noise to a random picture. This will be done by constructing an algorithm which will be implemented on the Mat-Lab for the graphical representation. The algorithm for the introduction of noise consists of a particular type of noise which can indicate a specific type of distortion.

Second step, demonstrated de-noise the image by introducing different filters. These filters functions to remove the distortion created in the first step of our study. These filters are constructed with the help of the algorithm. Later, it was implemented with the Mat-Lab for the graphical representation as in the first step. These filters are studied and discussed in the literature review. Output image was obtained in third step, Furthermore; this information is then applied during the experiments and achieved the desirable results i.e. gaining the actual image again adding and removing noise.

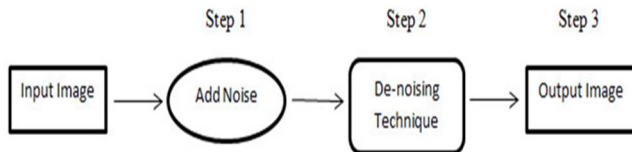


Fig. 2. Flowchart of De-noising Process

V. RESULTS AND DISCUSSION

We have implemented the different noise on our ground truth images and then later we have applied different filters in order to remove that noise from the images as shown in Fig. (5) (6) (7) (8).

In Fig. (5), we had applied salt and pepper noise on the Ground truth image. Fig. (5.b) clearly depicted the salt and noise small particles. We then applied different filters to remove salt and pepper noise such as Linear Filtered Image, FIR Filtered Image, Adaptive Filtered Image, and Median Filtered Image. Results clearly shows that Median Filter Image is more clear result than other filters. Median filter also preserve the edges.

In Fig. (6), (7), (8) we had applied different types of noise on the Ground truth image. Fig. (6.b), (7.b), (8.b) clearly depicted the impact of different noise. We then applied different filters to different noise such as Linear Filtered Image, FIR Filtered Image, Adaptive Filtered Image, and Median Filtered Image. Results clearly shows that Adaptive Filter Image has better results.

We also check the MSE and CPSNR performance for four filters.

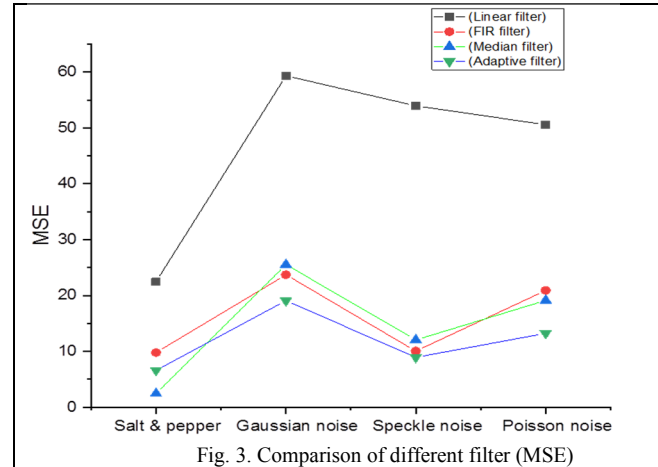


Fig. 3. Comparison of different filter (MSE)

$$MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

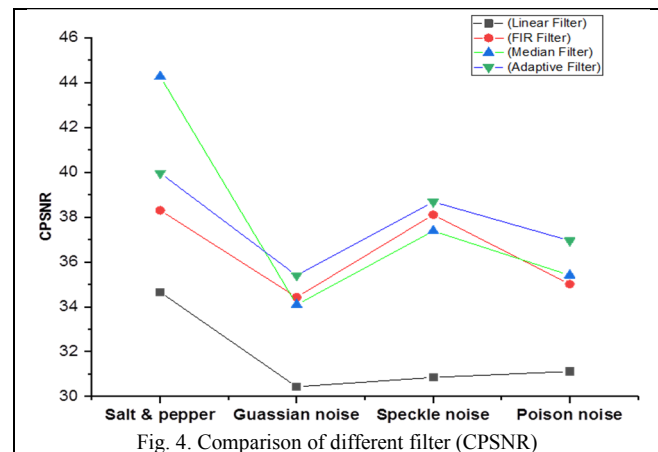


Fig. 4. Comparison of different filter (CPSNR)

$$CPSNR = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE} \right)$$

Table 1. Shows the MSE for the R, G and B component of four different images (Kodak dataset) on four different types of noise. The lowest MSE in each row is highlighted in bold. The Adaptive filter give the lowest MSE for all images except for first image, median filter gave lowest MSE. Adaptive filter (wiener filter) reduces MSE for Gaussian, poison and speckle noise.

Table 2. Shows us the CPSNR for the R, G and B component of four different images (Kodak dataset) on four different types of noise. The highest CPSNR in each is highlighted in bold. Median filter give highest CPSNR for salt and pepper noise, while Adaptive filter (wiener filter)

give highest CPSNR for other three noise. The experiment result shows us that Adaptive filter (wiener filter) has excellent output results as objectively and subjectively image quality.

Table 1. MSE (R, G, B) values calculations

Noise Type	Linear Filtered		FIR Filtered		Median Filtered		Adaptive Filtered	
Salt and pepper noise	R	22.31	R	10.83	R	2.32	R	6.81
	G	22.76	G	10.65	G	2.48	G	6.75
	B	22.31	B	7.89	B	2.56	B	6.25
	Av g.	22.46	Av g.	9.79	Av g.	2.45	Av g.	6.60
Gaussian noise	R	61.25	R	23.07	R	25.97	R	18.03
	G	56.42	G	26.74	G	24.97	G	23.08
	B	60.31	B	21.33	B	25.65	B	16.12
	Av g.	59.32	Av g.	23.71	Av g.	25.53	Av g.	19.07
Speckle noise	R	57.36	R	11.03	R	13.53	R	9.87
	G	56.02	G	10.44	G	12.55	G	9.47
	B	48.45	B	8.64	B	10.06	B	7.44
	Av g.	53.94	Av g.	10.03	Av g.	12.04	Av g.	8.92
Poisson noise	R	49.40	R	23.79	R	22.10	R	13.24
	G	50.04	G	22.29	G	20.40	G	13.35
	B	52.31	B	16.64	B	14.85	B	13.09
	Av g.	50.58	Av g.	20.90	Av g.	19.11	Av g.	13.22

TABLE 2. CPSNR(R, G, B) values calculations for different filter performance (in DB)

Noise Type	Linear Filtered		FIR Filtered		Median Filtered		Adaptive Filtered	
Salt and pepper noise	R	34.68	R	37.82	R	44.50	R	39.83
	G	34.59	G	37.89	G	44.22	G	39.86
	B	34.68	B	39.19	B	44.09	B	40.20
	Av g.	34.65	Av g.	38.30	Av g.	44.27	Av g.	39.96
Gaussian noise	R	30.29	R	34.53	R	34.01	R	35.60
	G	30.65	G	33.89	G	34.19	G	34.5
	B	30.36	B	34.87	B	34.07	B	36.09
	Av g.	30.43	Av g.	34.43	Av g.	34.09	Av g.	35.40
Speckle noise	R	30.57	R	37.73	R	36.85	R	38.22
	G	30.68	G	37.97	G	37.17	G	38.39
	B	31.31	B	38.79	B	38.13	B	39.44
	Av g.	30.85	Av g.	38.10	Av g.	37.38	Av g.	38.68
Poisson noise	R	31.22	R	34.40	R	34.72	R	36.94
	G	31.17	G	34.68	G	35.06	G	36.91
	B	30.97	B	35.95	B	36.44	B	36.99
	Av g.	31.12	Av g.	35.01	Av g.	35.41	Av g.	36.95

VI. CONCLUSION & FUTURE WORKS

This paper investigates the noise in the image along with some de-noising techniques, with comprehensive literature review which have added adequate knowledge related to the field of image noise processing. Our objective was to have a collective literature review along with some experiment to have the insight knowledge. Our limitation will include the number of filters and four different types of noise, which we have used in our experiments. The results show that in the presence of different kinds of noises Adaptive filter provide very clear and robust performance.

Future work will include our designed filter to deal with such noises; also we would like to incorporate more noise to our next experiments. Furthermore, we would like to overcome our limitations by incorporating different filters that are more effective for image de-noising. Our future task will also focus on the creation of a filter that can encounter different noise at the same session and produce better results while removing noise from the image.

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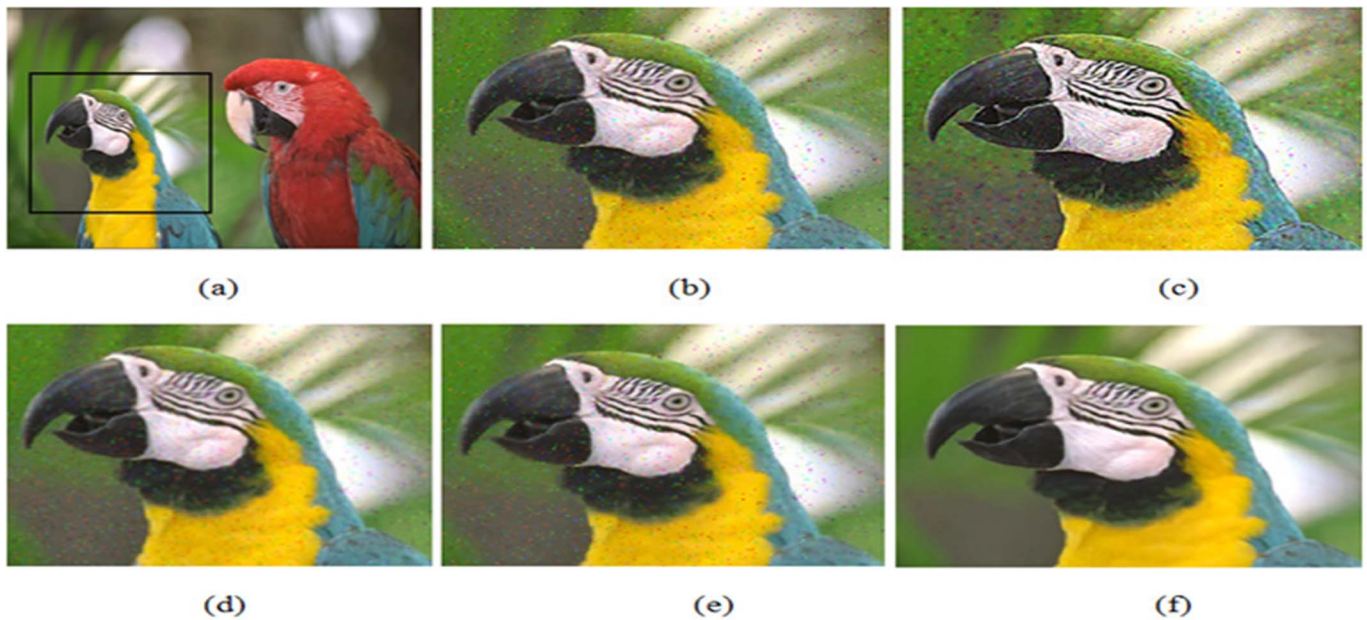


Fig. 5. a) Ground truth image b) Salt and pepper noise c) Linear Filtered Image d) FIR Filtered Image e) Adaptive Filtered Image f) Median Filtered Image.

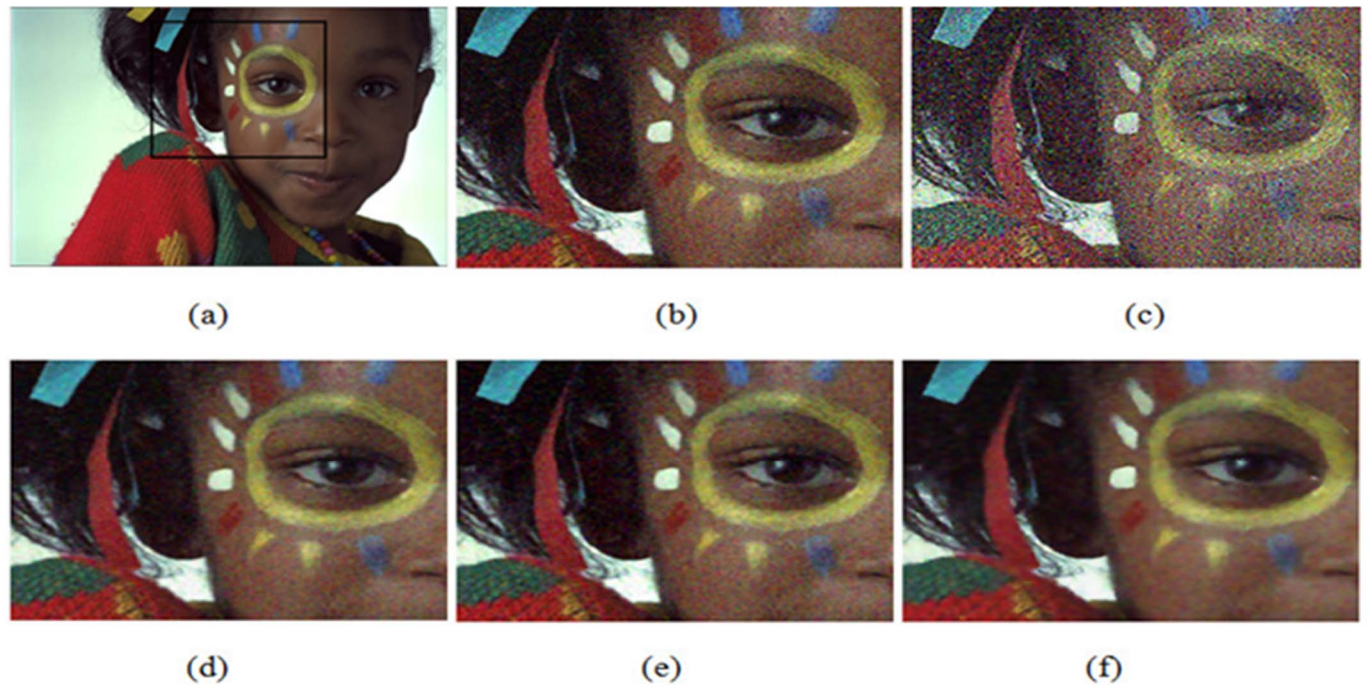


Fig. 6. a) Ground truth image b) Gaussian noise c) Linear Filtered Image d) FIR Filtered Image e) Median Filtered Image f) Adaptive Filtered Image.

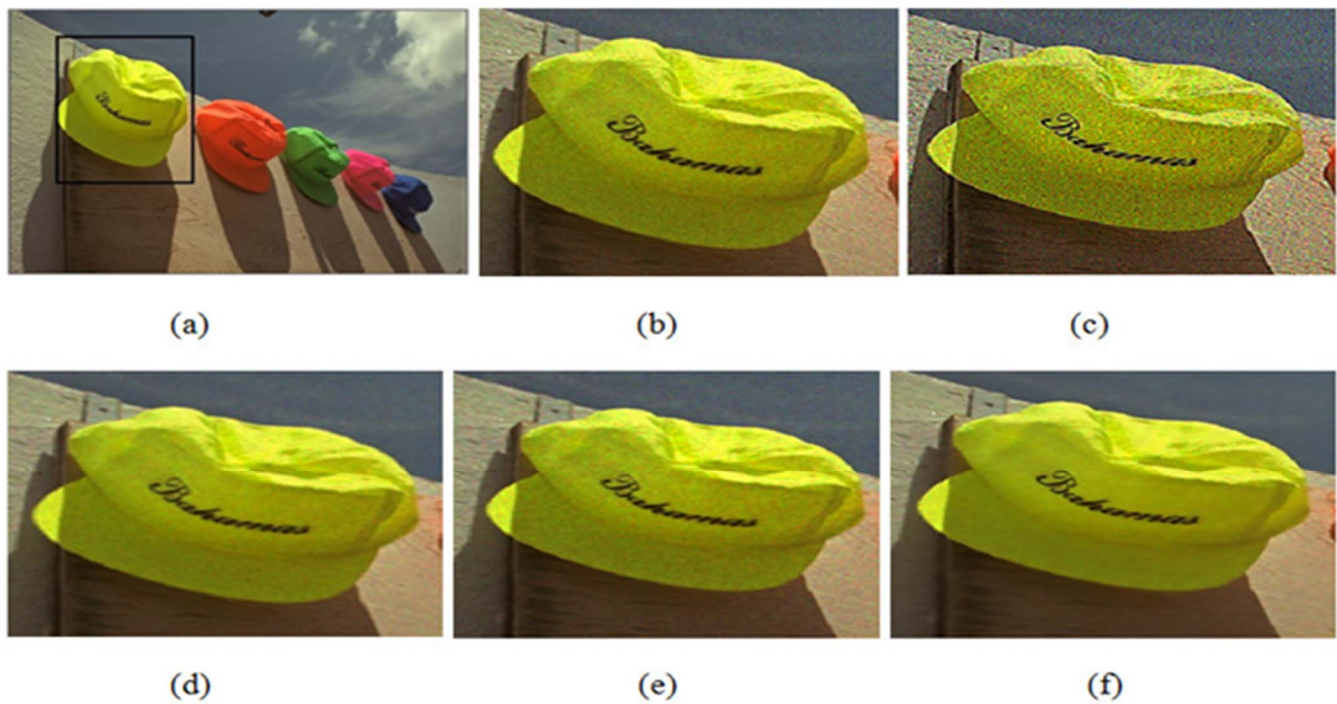


Fig. 7. a) Ground truth image b) Speckle noise c) Linear Filtered Image d) FIR Filtered Image e) Median Filtered Image f) Adaptive Filtered Image.

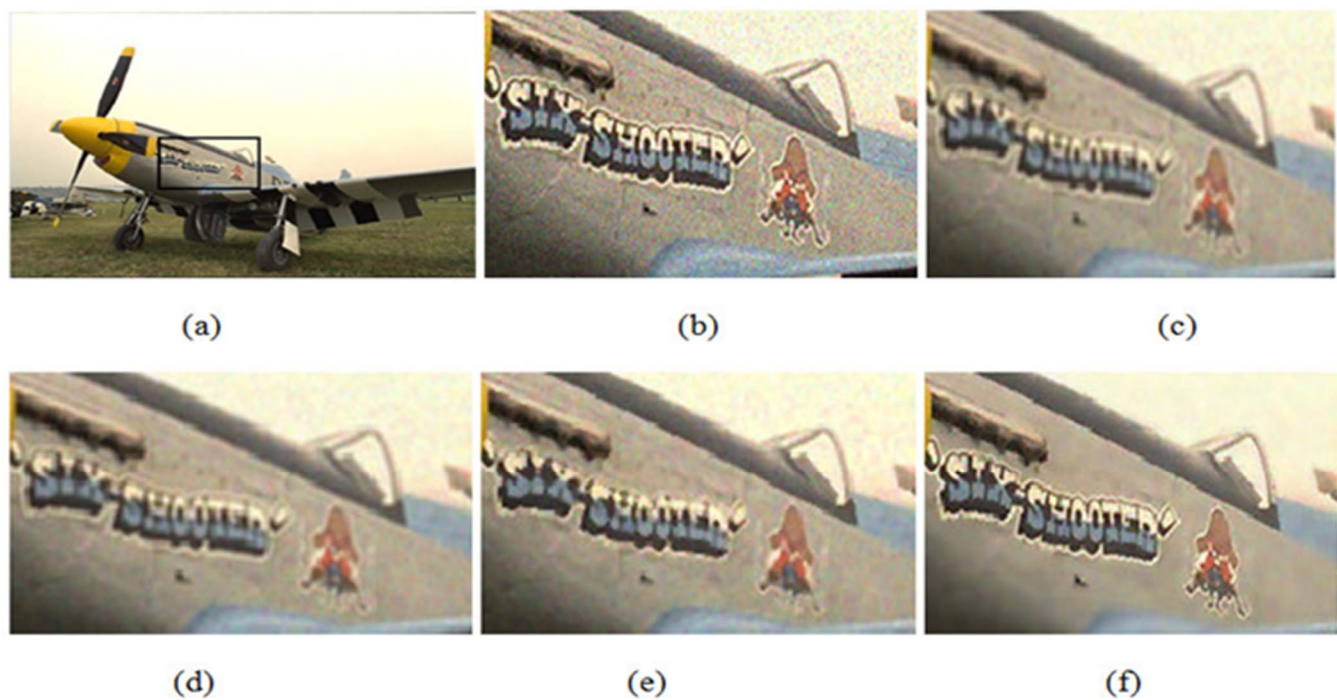


Fig. 8. a) Ground truth image b) Poisson noise c) Linear Filtered Image d) FIR Filtered Image e) Median Filtered Image f) Adaptive Filtered Image.